Progress Report

Applying the CMM-VAE Method to MSWEP Precipitation Data over Morocco

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Data Exploration

I started by exploring the new **MSWEP (Multi-Source Weighted-Ensemble Precipitation)** dataset, which was provided as a replacement for the CHIRPS dataset. The data is stored in NetCDF format and contains daily precipitation values in millimetres per day. During exploration, I found the following key characteristics:

- The precipitation variable is named `precipitation` (different from CHIRPS, which used `precipitation_amount`)
- The coordinates are `latitude`, `longitude`, and `time`
- The dataset covers a global domain, so a geographic filter was needed to extract Morocco specifically
- The available time period is **1979 to 2020**, which I aligned with the ERA5 Z500 predictor data to ensure both datasets cover the same period
- The spatial resolution is **0.1° × 0.1°**, providing a fine grid over Morocco
- Only **extended winter months** (November, December, January, February, March) were selected, as these are the dominant rainfall months in Morocco

Generating the Target Files from MSWEP

Before training the model, I processed the raw MSWEP data to generate the three files required by the CMM-VAE. This was done using the script `generate_mswep_files.py`.

Step 1 — Geographic Filter

I applied a strict bounding box covering the **whole of Morocco**: latitudes from 21°N to 36°N and longitudes from 17°W to 1°W. This ensures only land areas relevant to Morocco are included, and that ocean data does not distort the clustering results.

Step 2 — K-Means Clustering (K = 5)

I applied K-Means clustering to the filtered precipitation fields to group all **6,353 extended winter days** into 5 precipitation regimes. These range from Regime 0 (anomalously wet days) to Regime 4 (dry days). The result was saved to `MSWEP_pr_cluster_labels_5.csv`.

Step 3 — 95th Percentile Exceedance Mask

I computed the 95th percentile precipitation threshold at each grid point and created a binary mask recording whether each day exceeded this threshold. This was saved to `MSWEP_pr_95pc_EW.nc`.

Step 4 — Total Precipitation Field

I saved the full spatial precipitation field for all winter days to `MSWEP_pr_total_EW.nc`, used later for cluster reordering and visualisation.

Training the CMM-VAE Model

I trained the model using the notebook `cmmvae_mswep.ipynb`.

What the Model Does

The CMM-VAE is a variational autoencoder that jointly models the Z500 atmospheric circulation patterns and the surface precipitation regimes. It consists of:

- **Encoder:** Reads the Z500 map and compresses it into a latent space. It also outputs a probability distribution over weather clusters (`c`) and directly predicts which precipitation regime is most likely (`r`).
- **Decoder:** Reconstructs the original Z500 map from the latent representation.

The key hyperparameters used are summarised in Table 1.

Table 1: CMM-VAE Hyperparameters

Parameter	**Value**
Z500 clusters (K)	5
Precipitation clusters	5
Epochs per fold	100
Batch size	128
Latent dimension	15
Encoder layers	$256 \rightarrow 128 \rightarrow 64$
Decoder layers	$64 \rightarrow 128 \rightarrow 256$
Optimiser	Adam

Training Strategy — 5-Fold Cross-Validation

I used **5-Fold Cross-Validation** to train the model robustly:

1. I divided all 6,353 winter days into 5 equal groups (folds)
2. In each round, I trained on 4 folds (100 epochs) and validated on the 5th fold
3. I repeated this 5 times so every day was used as a validation sample once
4. The fold with the lowest final validation loss was selected as the final model (Fold 5, validation loss = 52.79)

5. Each fold started from the same initial random weights to ensure a fair comparison

Testing Methodology

Out-of-Fold Predictions (Brier Skill Score)

During each fold of the K-Fold training, the model made predictions on the held-out validation days — days it had **never trained on**. I collected all these predictions into a variable called `out_of_fold_predictions`. By the end of all 5 folds, every single day in the dataset had been predicted exactly once as unseen data. I then evaluated these predictions using the **Brier Skill Score (BSS)**, which measures how much better the model is compared to simply predicting the historical average every day:

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}_{\mathrm{model}}}{\mathrm{BS}_{\mathrm{climatology}}} \tag{1}$$

A BSS of 0 means no improvement over climatology. A positive BSS means the model adds genuine predictive value.

Direct Accuracy Check

I also tested the model's direct precipitation prediction output (`r`) against the true K-Means cluster labels. This gives a simple accuracy percentage: how often the model picks the correct precipitation regime.

Results

Brier Skill Scores

Table 2 shows the Brier Skill Scores computed on the out-of-fold predictions, i.e., on strictly unseen data. All three scores are positive, meaning the model consistently out-

Table 2: Brier Skill Scores on out-of-fold (unseen) data

Metric	BSS Score	Meaning
BSS — 95th Percentile	0.164	16% better than climatology for extreme rain days
BSS — Cluster	0.175	Correctly identifies precipitation regime above chance
BSS — Tercile (mean)	0.069	Modest skill at predicting above/below average rain

performs the climatological baseline on days it had never seen during training.

Precipitation Cluster Accuracy

Using the model's direct precipitation output (`r`), I compared predictions against the true cluster labels for all 6,353 days:

Table 3: Precipitation cluster accuracy

Metric	Value
Overall Accuracy	**92.3%**
Correct predictions	5,866 / 6,353 days
Random baseline	20.0% (1 in 5)

Z500 Cluster Centres

Figure 1 shows the composite Z500 anomaly patterns associated with each of the 5 identified weather regimes, ordered from the wettest (Regime 1) to the driest (Regime 5) over Morocco.

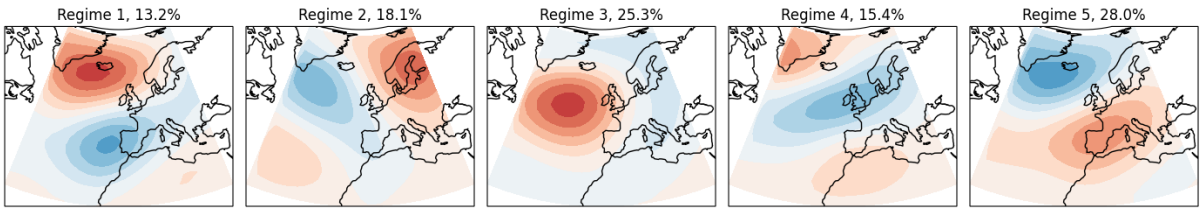

Figure 1: Z500 composite anomaly maps for the 5 weather regimes identified by the CMM-VAE, ordered by total precipitation over Morocco (wettest to driest).

Precipitation Odds Ratio

Figure 2 shows the odds ratio of exceeding the 95th percentile precipitation threshold over Morocco for each of the 5 regimes. Values greater than 1 indicate a higher-than-average probability of extreme precipitation during that regime.

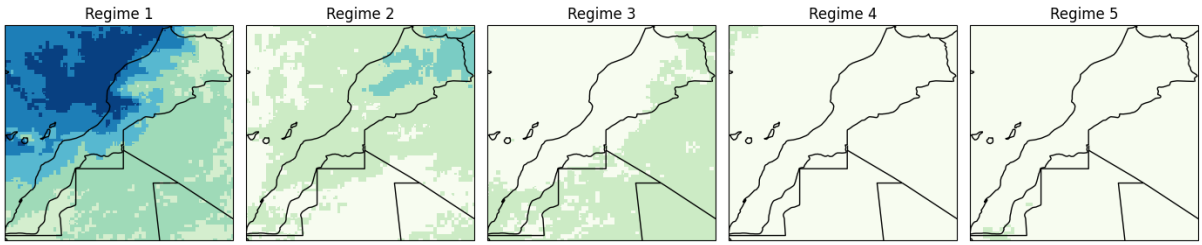

Figure 2: Odds ratio of 95th percentile precipitation exceedance over Morocco for each of the 5 weather regimes identified by the CMM-VAE (MSWEP dataset, 1979–2020).